Practice workbook · Where two lines meet

Finding the intersection of two straight lines by solving their equations simultaneously, sometimes with a letter left in the answer.

4 question parts: 1 worked-example retry and 3 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-COO-07

1. Question 2 · part b · 3 marks

2023 JAN · Paper 01

Worked-example retry · Attempt part b only.

2 The point A has coordinates $(-5, 3)$, the point B has coordinates $(4, 0)$ and the point C has coordinates $(-1, 5)$.

The line l passes through C and is perpendicular to AB.

(a) Find an equation of l.
Give your answer in the form $ax + by + c = 0$ where a, b and c are integers.

(4)

The line l intersects AB at the point D.

(b) Show that the coordinates of D are $(-2, 2)$.

(3)

(c) Show that l is not the perpendicular bisector of AB.

(2)

(d) Find the value of $\tan \angle ABC$.
Give your answer in its simplest form.

(4)

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1. Question 2 · part b · 3 marks · Working and answer

Working and answer

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2. Question 11 · part d · 3 marks

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part d only.

11 The points A and B have coordinates $(-1, 3)$ and $(5, 6)$ respectively.

(a) Find an equation for the line AB.

(2)

The point P divides AB in the ratio $2 : 1$

(b) Show that the coordinates of P are $(3, 5)$

(2)

The point C with coordinates (m, n), where $m > 0$, is such that CP is perpendicular to the line AB.

Given that the radius of the circle which passes through A, P and C is 5

(c) find the value of m and the value of n.

(6)

The point D with coordinates (p, q) is such that the line AD is perpendicular to the line AB and the line DC is parallel to the line AB.

(d) Find the value of p and the value of q.

(3)

(e) Find the area of trapezium $ABCD$.

(4)

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2. Question 11 · part d · 3 marks · Working and answer

Working and answer

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3. Question 6 · part b · 5 marks

2020 JAN · Paper 01

Further practice · Attempt part b only.

6 The point A has coordinates (3, 0) and the point B has coordinates (2, 2).
The line L_1 passes through B and is perpendicular to AB.

(a) Find an equation of L_1

Give your answer in the form $ax + by + c = 0$

(5)

The line L_2 with equation $x - 7y - 3 = 0$ intersects the line L_1 at the point C.
The midpoint of AC is M.

(b) Find the coordinates of M.

(5)

(c) Find the area of the triangle ABM.

(4)

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3. Question 6 · part b · 5 marks · Working and answer

Working and answer

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4. Question 9 · part a · 7 marks

2022 JUN · Paper 02

Further practice · Attempt part a only.

9 The straight line L_1 passes through the point A with coordinates (4, 7) and has gradient m, where $m < 0$

Another straight line L_2 is perpendicular to L_1 and passes through the point B with coordinates $(4, k)$ where $k \neq 7$

The lines L_1 and L_2 intersect at the point C.

Given that the y coordinate of C is Y

(a) show that $Y = \dfrac{7 + m^2k}{m^2 + 1}$

(7)

Given that the triangle ABC is isosceles,

(b) find the value of m

(5)

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4. Question 9 · part a · 7 marks · Working and answer

Working and answer

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